

AUGUST/RESIT EXAMINATIONS 2022/2023

MODULE: CA266 - Probability & Statistics

PROGRAMME(S):

COMSCI	BSc in Computer Science
CASE	BSc in Computer Applications (Sft.Eng.)
ECSAO	Study Abroad (Engineering & Computing)

YEAR OF STUDY: 2,0

EXAMINER(S):

Dr. Graham Healy	(Internal)	(Ext:7364)
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TIME ALLOWED: 2 Hours

INSTRUCTIONS: Answer 4 questions. All questions carry equal marks.

PLEASE DO NOT TURN OVER THIS PAGE UNTIL YOU ARE INSTRUCTED TO DO SO.

The use of programmable or text storing calculators is expressly forbidden.

Please note that where a candidate answers more than the required number of questions, the examiner will mark all questions attempted and then select the highest scoring ones.

There are no additional requirements for this paper.

QUESTION 1**[TOTAL MARKS: 25]**

For each of these questions **explain your answer** i.e., show your reasoning.

Q 1(a)**[13 Marks]**

A computer system uses passwords that have five characters, and each character is one of the 26 letters (a-z) or 10 integers (0-9). The first character has to be a letter. You may assume that passwords are not case sensitive.

Determine the proportion of passwords that:

- i. end with an even number (0,2,4,6,8)
- ii. begin with a consonant **and** end with an even number
- iii. begin with a consonant **or** end with an even number **or** both
- iv. read the same backwards as forwards

Q 1(b)**[12 Marks]**

Suppose you have a box that contains 3 red balls, 3 blue balls, 2 yellow balls, 2 purple and 1 green ball. Suppose we were to sample **without replacement** from the box and draw N balls.

- i. What is the probability that the second ball is green given that the first ball was green?
- ii. What is the probability that the green ball is the 3rd ball drawn?
- iii. If I draw 4 balls, what is the probability that none of these will be red or blue?
- iv. If I draw 3 balls, what is the probability that I will observe the following sequence of ball colours: red, green, red?

[End of Question 1]

QUESTION 2**[TOTAL MARKS: 25]**

The table below shows the number of machine failures for the past 12 months (52 weeks) for a company. For example, for 9 of the weeks there were 0 failures, for 14 of the weeks there was 1 failure, and so on.

You can assume failures are independent of each other.

Number of failures	0	1	2	3	4	5	6
Number of weeks	9	14	13	9	4	2	1

For each of these questions **explain your answer** i.e., show your reasoning.

Q 2(a)**[10 Marks]**

What are the mean and variance for the number of failures per week?

Q 2(b)**[8 Marks]**

Give values for the following probabilities:

- $P(\text{the number of failures in a week will be equal to } 2)$
- $P(\text{the number of failures in a week will be less than or equal to } 2)$
- $P(\text{the number of failures in a week will be equal to } 2 \text{ given that } 1 \text{ failure has already occurred that week})$

Q 2(c)**[7 Marks]**

The company uses the following formula to calculate the monetary cost in Euro for machine failures in a week:

$$\text{Cost} = 5X + 50$$

Where X = number of failures

- What is the expected mean cost of failures in a week? i.e. find $E(5X + 50)$
- What is the expected variance of cost of failures in a week? i.e. find $V(5X + 50)$

[End of Question 2]

QUESTION 3**[TOTAL MARKS: 25]**

Suppose cars on a production line have a 0.9 probability of being defective. The test used to determine if a car is defective or not is not always correct. If a car is defective, there is a 0.2 probability that the test will show that the car is not defective (negative test). If the car is not defective, there is a 0.1 probability that the test will show the car is defective (positive test).

For each of these questions **explain your answer** i.e., show your reasoning.

Q 3(a)**[3 Marks]**

What is the probability that a car is not defective?

Q 3(b)**[3 Marks]**

What is the probability of a positive test given that a car is defective?

Q 3(c)**[6 Marks]**

What is the probability that a test is positive?

Q 3(d)**[7 Marks]**

What is the probability that a car is not defective, given that a test is positive?

Q 3(e)**[6 Marks]**

What is the probability that if I select 5 random cars from the production line, that at least two of these will be defective?

[End of Question 3]

QUESTION 4**[TOTAL MARKS: 25]**

Suppose there are 2 car production lines in a factory that operate 24 hours a day, 7 days a week, 365 days a year. Both production lines are identical. Line A produces on average 2 cars a day, and line B also produces on average 2 cars a day.

Suppose that the number of cars produced on each production line obey a Poisson Distribution.

For each of these questions **explain your answer** i.e., show your reasoning.

Q 4(a) **[4 Marks]**

What is the probability that line A will produce 0 cars on a given day?

Q 4(b) **[5 Marks]**

What is the probability that **both** line A and line B will produce 1 or more cars **each** on a given day?

Q 4(c) **[5 Marks]**

What is the probability that in a 12-hour period, line A or line B (or both) will produce 0 cars?

Q 4(d) **[6 Marks]**

What is the probability that both production lines will produce the same number of cars on a given day? You can give your answer correct to 2 decimal places.

Q 4(e) **[5 Marks]**

For what duration of time (in hours) would the probability of line A generating no cars be 0.61?

[End of Question 4]

QUESTION 5**[TOTAL MARKS: 25]**

Answer any 2 of the following questions i.e. of Q5(a), Q5(b) or Q5(c). There are equal marks for both questions.

For each answered question **explain your answers** i.e. show your reasoning.

Q 5(a)**[12.5 Marks]**

Machine A and machine B produce the same type of component. We are told the mean length of components for both machines is 5.2cm. The standard deviation for components produced by machine A is 1 cm. We are not told the standard deviation for components produced by machine B. You can assume component lengths for both machines are normally distributed.

- i. What is the probability that a component produced by machine A has a length greater 5.7cm?
- ii. What is the probability that a component produced by machine A has a length greater than 5.7cm or less than 4.7cm?
- iii. If we are told there is a 0.242 probability that a component produced by machine B is greater than 6.2 cm, what is the standard deviation of the component lengths produced by machine B?

Q 5(b)**[12.5 Marks]**

If you throw two dice, let the total score be X .

- I. What is $P(\text{both of the scores are the same})$?
- II. What are the probabilities for each value of X ?
- III. What is $P(\text{one of the scores is } 6 \mid X = 10)$?

Q 5(c)**[12.5 Marks]**

A box contains 20 apples. We are told 25% of the apples in the box are rotten.

- i. If 4 apples are taken from the box without replacement, what is the probability all 4 apples are rotten?
- ii. If 5 apples are taken from the box without replacement, what is the probability that 2 or more of these apples are rotten?
- iii. If the box of apples was infinite in size, how would this change how we calculate part i and part ii above? Explain your answer.

[End of Question 5]

APPENDICES

Area under the standard normal curve from $-\infty$ to z



z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
0.0	0.5000	0.5040	0.5080	0.5120	0.5160	0.5199	0.5239	0.5279	0.5319	0.5359
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.5753
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141
0.3	0.6179	0.6217	0.6255	0.6293	0.6338	0.6368	0.6406	0.6443	0.6480	0.6517
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.6808	0.6844	0.6879
0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157	0.7190	0.7224
0.6	0.7257	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454	0.7486	0.7517	0.7549
0.7	0.7580	0.7611	0.7642	0.7673	0.7704	0.7734	0.7764	0.7794	0.7823	0.7852
0.8	0.7881	0.7910	0.7939	0.7967	0.7995	0.8023	0.8051	0.8078	0.8106	0.8133
0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.8340	0.8365	0.8389
1.0	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554	0.8577	0.8599	0.8621
1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.8770	0.8790	0.8810	0.8830
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980	0.8997	0.9015
1.3	0.9032	0.9049	0.9066	0.9082	0.9099	0.9115	0.9131	0.9147	0.9162	0.9177
1.4	0.9192	0.9207	0.9222	0.9236	0.9251	0.9265	0.9279	0.9292	0.9306	0.9319
1.5	0.9332	0.9345	0.9357	0.9370	0.9382	0.9394	0.9406	0.9418	0.9429	0.9441
1.6	0.9452	0.9463	0.9474	0.9484	0.9495	0.9505	0.9515	0.9525	0.9535	0.9545
1.7	0.9554	0.9564	0.9573	0.9582	0.9591	0.9599	0.9608	0.9616	0.9625	0.9633
1.8	0.9641	0.9649	0.9656	0.9664	0.9671	0.9678	0.9686	0.9693	0.9699	0.9706
1.9	0.9713	0.9719	0.9726	0.9732	0.9738	0.9744	0.9750	0.9756	0.9761	0.9767
2.0	0.9772	0.9778	0.9783	0.9788	0.9793	0.9798	0.9803	0.9808	0.9812	0.9817
2.1	0.9821	0.9826	0.9830	0.9834	0.9838	0.9842	0.9846	0.9850	0.9854	0.9857
2.2	0.9861	0.9864	0.9868	0.9871	0.9875	0.9878	0.9881	0.9884	0.9887	0.9890
2.3	0.9893	0.9896	0.9898	0.9901	0.9904	0.9906	0.9909	0.9911	0.9913	0.9916
2.4	0.9918	0.9920	0.9922	0.9925	0.9927	0.9929	0.9931	0.9932	0.9934	0.9936
2.5	0.9938	0.9940	0.9941	0.9943	0.9945	0.9946	0.9948	0.9949	0.9951	0.9952
2.6	0.9953	0.9955	0.9956	0.9957	0.9959	0.9960	0.9961	0.9962	0.9963	0.9964
2.7	0.9965	0.9966	0.9967	0.9968	0.9969	0.9970	0.9971	0.9972	0.9973	0.9974
2.8	0.9974	0.9975	0.9976	0.9977	0.9977	0.9978	0.9979	0.9979	0.9980	0.9981
2.9	0.9981	0.9982	0.9982	0.9983	0.9984	0.9984	0.9985	0.9985	0.9986	0.9986
3.0	0.9987	0.9987	0.9987	0.9988	0.9988	0.9989	0.9989	0.9989	0.9990	0.9990

${}^nC_k = \frac{{}^nP_k}{k!} = \frac{n!}{k!(n-k)!}$	
${}^nP_k = n(n-1)\dots(n-k+1) = \frac{n!}{(n-k)!}$	
$P(E_1 \cup E_2) = P(E_1) + P(E_2) - P(E_1 \cap E_2)$	
$P(A B) = \frac{P(A)P(B A)}{P(B)}$	$P(A \cap B) = \frac{P(A)P(B A)}{P(B)P(A B)}$
$V(X) = \sum_x (x - \mu_x)^2 p(x)$	$E(X) = \sum_x xp(x)$
$P(X = x) = \binom{n}{x} p^x q^{n-x}$	$P(X \leq x) = 1 - q^x$ $P(X = x) = q^{x-1} p$
$P(X = x) = \frac{\binom{M}{x} \binom{L}{n-x}}{\binom{N}{n}}$	$P(X = x) = e^{-\lambda} \frac{\lambda^x}{x!}$

[END OF APPENDICES]

[END OF EXAM]